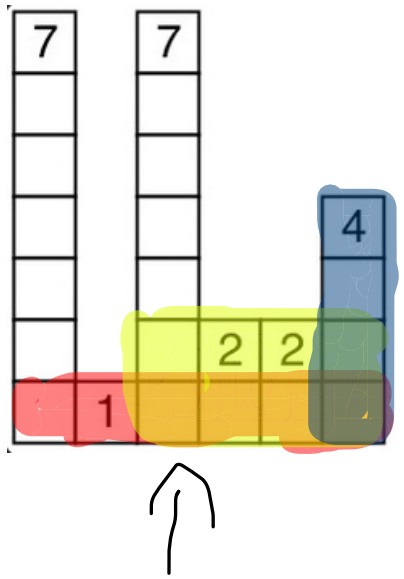


Stack

Pos	Height
0	7
2	4
3	7
4	2
5	1
6	4

1 is smaller than 7 so the 7 won't be contributing any more to the area (it can't be extended forward anymore, meaning there are no numbers bigger than it RIGHT after it)
so we first calculate its area see if it's bigger than the max and remove it
why 1 takes the position 0 not 1 is because it can be extended backwards , if position 1 , then its area is 1, but position 0 , its area is 2 , and it's objectively better (no reason to not use 0)

same thing with 2 < 7
we pop 7 , and since 2 can be extended backwards , it takes the popped index as its starting position and the max area is 7 and the area that the second 7 provides is 7 so no changes

Now the ones left in the stack are the ones that can be extended until the end of the histogram , so we can just calculate their maximum areas with : (histogram size - position) * height and compare to the maxArea , if there is one bigger , replace the maxArea with it

maxArea

46 x 8 ✓ 6 x 4 x